

SAHIL BHOIR

Software Engineer | Applied Machine Learning | Backend and AWS

Thane, Maharashtra | [Email](#) | [LinkedIn](#) | [GitHub](#) | [Portfolio](#) | [Medium](#)

SUMMARY

I am a Computer Engineering graduate focused on applied machine learning, backend engineering and AWS. I build complete systems around models - from data preparation and evaluation to APIs, cloud workflows, testing and user-facing interfaces - while keeping results and technical trade-offs defensible.

PROJECTS

Cyberattack Tactics Triage | [Python](#) · [TF-IDF](#) · [Linear SVM](#) · [Calibration](#) · [MITRE ATT&CK](#) | [GitHub](#)

- **Achieved 0.773 macro F1 and 0.826 micro F1** by training a multilabel Linear SVM on 16,955 cleaned Enterprise ATT&CK procedure examples spanning 15 tactics.
- **Strengthened evaluation reliability** by replacing a random split with source-grouped partitions and removing eight cross-partition near-duplicates.
- **Built an analyst-review workflow** with calibrated confidence, feature-level explanations and CSV batch processing; an unseen-technique stress test recorded 0.516 macro F1, exposing the boundary on unfamiliar behaviour.

Real-Time Violence Event Detection | [YOLOv8](#) · [OpenCV](#) · [FastAPI](#) · [Streamlit](#) · [Telegram](#) | [Demo](#) · [GitHub](#)

- **Evaluated 17,289 frame observations across 35 reviewed clips** to select temporal settings for violent, nonviolent and hard-negative scenes.
- **Designed an N-positive/K-negative state machine** that converted noisy pretrained YOLOv8 frame predictions into persistent events with controlled activation and release.
- **Delivered an end-to-end monitoring workflow** with video, webcam and RTSP inputs, FastAPI controls, Streamlit review, stored screenshots, runtime diagnostics and optional Telegram alerts; detector-level misses remained separate from temporal-filter performance.

Serverless Receipt Processor | [AWS Lambda](#) · [Cognito](#) · [S3](#) · [Step Functions](#) · [Textract](#) · [DynamoDB](#) | [Demo](#) · [GitHub](#)

- **Built an authenticated serverless workflow** that transformed uploaded receipts into user-scoped, editable expense records using Cognito, S3, Textract, Lambda and DynamoDB.
- **Separated upload and processing workloads** through presigned S3 URLs, EventBridge and Step Functions, keeping OCR and parsing outside the client request path.
- **Improved extraction reliability** with tested rules for ambiguous totals and Indian currency and date formats, while retaining the original receipt and editable fields when OCR remained uncertain.

EDUCATION

Vidyalankar Institute of Technology | B.E. Computer Engineering, AI and ML Minor | **CGPA: 9.43/10** | 2021-2025

TECHNICAL SKILLS

Languages: Python, JavaScript, Java, C/C++

Machine Learning: PyTorch, scikit-learn, XGBoost, Pandas, NumPy, Transformers, OpenCV

Backend and Cloud: FastAPI, Flask, REST APIs, WebSockets, AWS Lambda, S3, DynamoDB, API Gateway, Step Functions

Engineering: Docker, Git/GitHub, GitHub Actions, Linux, automated testing

CERTIFICATIONS AND WRITING

Certifications: AWS Certified Cloud Practitioner (CLF-C02) · IBM Full Stack Software Developer Professional Certificate

Writing: Articles on AI systems, agents, work and computational mythology - [Medium profile](#)